

endpoint 狀態：名稱要和機制對齊

Context: local fallback result / endpoint boundary

Run: python -m json.tool fallback_artifacts/baseline-A/result.json

rg -n 'STATE_INTERVAL|state|duration' fallback_artifacts/baseline-A/result.json



每個 state 都留下 duration / power / event ; endpoint J 是累積結果。

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

WAIT 是 awake idle ， SLEEP 會帶來 wake 成本

Context: baseline-A endpoint replay ； 同一 job / window / energy boundary

Run: rg -n '"state": "WAIT"|"state": "SLEEP"|"state": "WAKE"' fallback_artifacts/baseline-A/result.json

WAIT

awake idle → no wake event

SLEEP

sleep → WAKE → service

先提出 WAIT 與 SLEEP 的方向性預測，再由 state ledger 驗證。 WAIT 保持清醒； SLEEP 可能降低 idle power，但後續 WAKE 會增加 latency / energy。

封包生命週期：送出不是交付

Context: baseline-A endpoint replay ; packet ledger

Run: python -m json.tool fallback_artifacts/baseline-A/endpoint-replay.json

queue

attempt

retry

delivered

expired

attempted $\neq$ delivered
retry / collision / expired 都要保留

一次 SEND 是 decision ; service
要看完整 packet outcome 。

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

service boundary 先於 energy 比較

Context: paired baseline / candidate ; 同 scenario 、 job 、 window 、 endpoint scope

Run: rg -n "service_pass"|"deadline_pass"|"energy_scope" fallback_artifacts/candidate-A/result.json

SERVICE GATE

same job
same window
delivered / deadline

THEN COMPARE J

endpoint scope
state ledger
energy breakdown

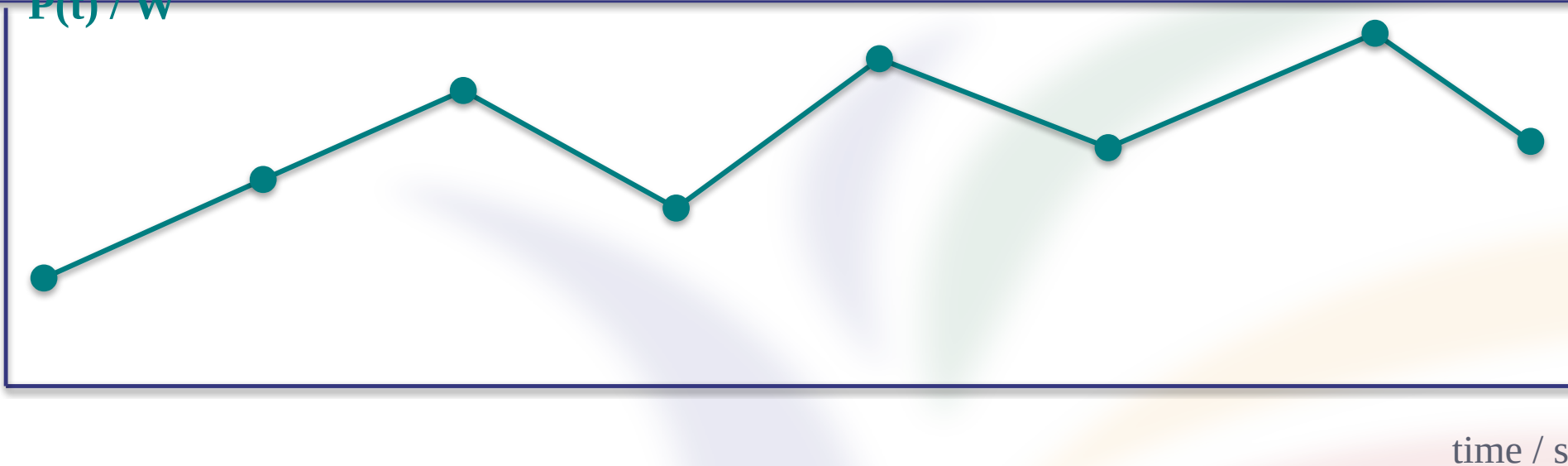
先確認 job 、 window 、 delivery 與 deadline 可比，再談 J 。 candidate 少送或多過期時，較低 J 不能直接升格為節能； service gate 先決定比較資格。

W 是速率，J 是累積結果

Context: result JSON ; energy scope = endpoint

Run: rg -n "endpoint_energy_j"|"power"|"energy_breakdown_j" fallback_artifacts/candidate-A/result.json

$P(t) / W$



曲線下的 area 是 endpoint J ；低峰值拖得久，仍可能累積更多。

公式： endpoint energy 的範圍

EDITABLE OFFICE MATH

$$E_{\text{endpoint}} = \sum P_s t_s$$

$E_{\text{endpoint}} = \sum P_s t_s$
endpoint boundary $\neq$ whole system

Office Math hook is editable ；數字仍由 result artifact 提供，不自行補 KPI 。

公式：服務效率的分子與分母

EDITABLE OFFICE MATH

$$\eta_E = D_{\text{delivered}} / E_{\text{endpoint}}$$

$$\eta_E = D_{\text{delivered}} / E_{\text{endpoint}}$$
$$\text{bit} \div J = \text{bit}/J$$

Office Math hook is editable ; 數字仍由 result artifact 提供，不自行補 KPI 。

來源、模型、課程假設、結果

Context: result / replay / scenario lineage

Run: rg -n 'scenario_id|seed|policy_sha256|artifact_source' fallback_artifacts/candidate-A/result.json

source

model

course

result

ntpu-energy-decision-01 · seed 12001 / 12002 · policy SHA · artifact_source

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

一個 scenario_id 綁住整條鏈

SOURCE

repo
[https://github.com/
cedarwud/lora-energy-
lab](https://github.com/cedarwud/lora-energy-lab)

commit
32cc72309041...

SCENARIO

ntpu-energy-decision-
01

anchor
sha256:cfe84682a71c0
...

lock
sha256:f1d1fe3339137
...

RUN → REPLAY → WORKBOOK

same scenario
same units
same seed role
policy SHA
fail closed on mismatch

IDENTITY GATE | scenario / anchor / policy / replay must agree

公平 baseline : 固定 job 、 window 、 boundary

Context: package root ; baseline / candidate A-B pair

Run: git diff -- student_policy.py

python -m py_compile student_policy.py

FIXED

scenario / seed / job / window

ONE EDIT

current marked block

OBSERVE

queue / packet / service / state
/ J

若 trace 、 scope 、 job 或 policy 偷換，這一組 A/B 就失去因果資格。

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

LEO window : 合法與不合法的 action

Context: scenario quality/contact trace ; same anchor

Run: rg -n 'contact|window|quality_band_trace' scenarios/ntpu-energy-decision-01.json

WINDOW OPEN

SEND / WAIT legal context

WINDOW CLOSED

delayed action → service risk

把 changing opportunity 接回 WAIT / SEND 的合法性。窗口開啟時 action 可能服務；窗口關閉時延後可能變成 service loss；LEO 只作 changing-window example。

Lab A : 同一份工作，不同 pace

LAB A QUESTION

same job
different pace / rest

先預測，再 baseline → candidate → withheld

WATCH THE CHAIN

policy edit
→ **state / transition**
→ **packet / service**
→ **endpoint J**

**FALLBACK ARTIFACTS ARE
SIMULATED**

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

先讀 PACE_GAP_STEPS

Context: package root ; student_policy.py 的 lab-a-pace-rest marked block
Run: sed -n '1,80p' student_policy.py
python -m py_compile student_policy.py

```
# === LORA EDITABLE: lab-a-pace-rest ===  
PACE_GAP_STEPS = 2  
REST_DURING_GAP = SLEEP  
# === LORA END EDITABLE ===
```

ONE BLOCK

gap → send opportunity
not direct energy
read steps_since_send

POLICY SURFACE | marked block only | syntax guard before run

再讀 REST_DURING_GAP

Context: package root ; student_policy.py 的 lab-a-pace-rest marked block
Run: sed -n '10,24p' student_policy.py
python -m py_compile student_policy.py

```
# === LORA EDITABLE: lab-a-pace-rest ===  
PACE_GAP_STEPS = 2  
REST_DURING_GAP = SLEEP  
# === LORA END EDITABLE ===
```

ONE BLOCK

WAIT stays awake
SLEEP may add wake cost
read state + service

POLICY SURFACE | marked block only | syntax guard before run

Lab A 的 prediction lock

Context: package root ; edit 前，保持 baseline policy

Run: git diff -- student_policy.py

python -m py_compile student_policy.py

queue age



/ hold — write a reason before run

service



/ hold — write a reason before run

state time



/ hold — write a reason before run

endpoint J



/ hold — write a reason before run

在 run 前鎖定 queue、service、state time、J 的判讀句。 prediction 先於 result；withheld 不能 retune 來迎合答案。

A-01 : 跑 untouched baseline

RUN THIS EXACTLY

POSIX / WSL ; package root lora-energy-lab/ ; baseline policy 不改
bash course.sh run --lab A --case baseline
Windows: course.cmd run --lab A --case baseline

Purpose: establish control.

Mechanism: fixed scenario / seed → result lineage.

READ THE RECEIPT

fallback 可讀 *run_id ...ee6df44f* 、 energy
6.92 J 、 delivered 4800 bit ; fresh
stdout/result path 仍待重跑。

IF THE GATE IS RED

保留 stdout/stderr ; restore A checkpoint
或 matching fallback ; 不要再調參。

VERIFIED LOCAL FALLBACK | SIMULATED | fresh receipt/path pending

A-02 : 只改 A marked block 並 freeze

RUN THIS EXACTLY

```
POSIX / WSL ; lora-energy-lab/ ; 只改 lab-a-  
pace-rest  
git diff -- student_policy.py  
bash course.sh run --lab A --case candidate --freeze  
# Windows: course.cmd run --lab A --case  
candidate --freeze  
Purpose: freeze one marked edit.  
Mechanism: policy SHA + predecessor + receipt.
```

READ THE RECEIPT

```
candidate policy 325cca4a...d73cf2f  
freeze 6cbaf9dc...e11626  
energy 8.86 J ; fresh receipt pending
```

IF THE GATE IS RED

保留 stdout/stderr ; restore A checkpoint
或 matching fallback ; 不要再調參。

VERIFIED LOCAL FALLBACK | SIMULATED | fresh receipt/path pending

A-03 : 匯入 candidate , 先看同一個 boundary

Context: local fallback pair ; result / endpoint-replay 同 scenario
Run: python -m json.tool fallback_artifacts/candidate-A/result.json
python -m json.tool fallback_artifacts/candidate-A/endpoint-replay.json

BASELINE

scenario ✓
policy sha256:9917d2cde2e...
energy 6.92 J · delivered 4800 bit
service_pass = false

CANDIDATE

scenario ✓
policy sha256:325cca4a4be...
energy 8.86 J · delivered 4800 bit
service_pass = false

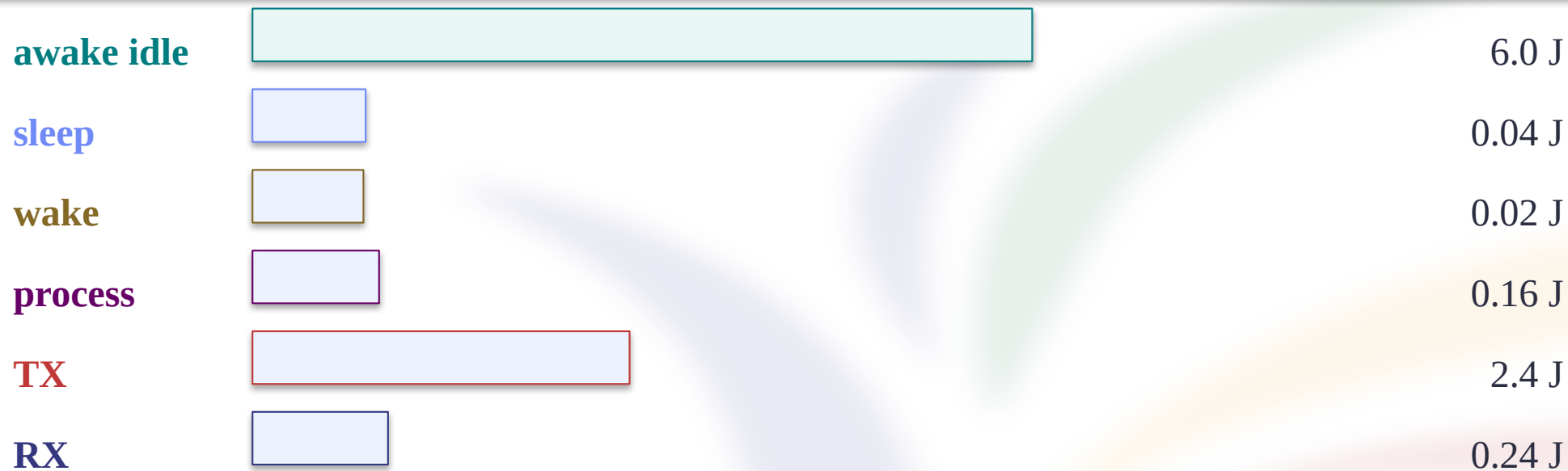
先確認同一 boundary ; candidate 的 trade-off 再交給 state / packet / service ledger 。

A-04 : 讀懂 state ledger

Context: candidate-A result ; state interval / energy breakdown

Run: rg -n 'STATE_INTERVAL|awake_idle|sleep|wake|process|tx|rx'

fallback_artifacts/candidate-A/result.json



先讀 state duration ，再問哪個 policy action 造成 consequential diff 。

A-05 : 把 packet 結果接回 service

Context: candidate-A result / endpoint replay ; packet ledger

Run: rg -n 'attempt|retry|delivered|expired|service_pass' fallback_artifacts/candidate-A/result.json

A CANDIDATE PACKET LEDGER

attempted 2
retransmissions 1
delivered 4800 bit
expired 3
energy 8.86 J

SERVICE VERDICT

service_pass = false
deadline_pass = false
先說 trade-off ，不說節能。

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

A-06 : 檢查 A freeze lineage

Context: package root ; A frozen receipt / checkpoint
Run: python -m json.tool fallback_artifacts/receipts/a.json
find artifacts/checkpoints -maxdepth 1 -type f -print

baseline

candidate

lab-a-frozen

policy SHA · predecessor · active_block_id · scenario · anchor · seed · receipt SHA

FREEZE GATE | missing receipt / checkpoint = stop before withheld

A-07 : 用 hidden case 檢驗，不再 retune

Context: POSIX / WSL ; A frozen policy active ; same package root
Run: bash course.sh run --lab A --case hidden
Windows: course.cmd run --lab A --case hidden

FROZEN POLICY

policy SHA unchanged
no new freeze
read result + replay

HIDDEN TRACE

new condition
queue / service / state / J
qualified verdict

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

Lab A debrief：一條因果句就夠

Context: A baseline / candidate / hidden fallback trio

Run: python -m json.tool fallback_artifacts/hidden-A/result.json

rg -n 'policy_sha256|service_pass|endpoint_energy_j' fallback_artifacts/{baseline-A,candidate-A,hidden-A}/result.json

condition

mechanism

evidence

verdict

在 Trace A / primary... ；在 Trace B / hidden... ；因此這個結論成立、被推翻，或待查。

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

Lab B：現在送，還是等到更穩定？

LAB B QUESTION

same job
quality trace / stability

先預測，再 baseline → candidate → withheld

WATCH THE CHAIN

policy edit
→ **state / transition**
→ **packet / service**
→ **endpoint J**

**FALLBACK ARTIFACTS ARE
SIMULATED**

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

ENTER_QUALITY : 何時進入 send-ready

Context: package root ; student_policy.py 的 lab-b-enter-exit-hold marked block
Run: sed -n '18,28p' student_policy.py
python -m py_compile student_policy.py

```
# === LORA EDITABLE: lab-b-enter-exit-hold ===  
ENTER_QUALITY = 2  
EXIT_QUALITY = 1  
STABLE_STEPS = 2  
# === LORA END EDITABLE ===
```

ONE BLOCK

quality + stable count
→ send-ready transition
not direct energy

POLICY SURFACE | marked block only | syntax guard before run

EXIT_QUALITY : 為什麼不能只用一條線

Context: package root ; student_policy.py 的 lab-b-enter-exit-hold marked block
Run: sed -n '18,28p' student_policy.py
python -m py_compile student_policy.py

```
# === LORA EDITABLE: lab-b-enter-exit-hold ===  
ENTER_QUALITY = 2  
EXIT_QUALITY = 1  
STABLE_STEPS = 2  
# === LORA END EDITABLE ===
```

ONE BLOCK

Purpose: separate enter and exit.
Mechanism: hysteresis limits ping-pong.

POLICY SURFACE | marked block only | syntax guard before run

STABLE_STEPS : 拒絕短暫尖峰

Context: package root ; student_policy.py 的 lab-b-enter-exit-hold marked block
Run: sed -n '18,28p' student_policy.py
python -m py_compile student_policy.py

```
# === LORA EDITABLE: lab-b-enter-exit-hold ===  
ENTER_QUALITY = 2  
EXIT_QUALITY = 1  
STABLE_STEPS = 2  
# === LORA END EDITABLE ===
```

ONE BLOCK

count stable steps
→ switch after hold
short spike rejected

POLICY SURFACE | marked block only | syntax guard before run

Lab B 的 Trace A prediction

Context: package root ; Trace A run 前 ; A frozen predecessor active

Run: git diff -- student_policy.py

python -m py_compile student_policy.py

queue age

↗ / ↘ / hold — write a reason before run

service

↗ / ↘ / hold — write a reason before run

state time

↗ / ↘ / hold — write a reason before run

endpoint J

↗ / ↘ / hold — write a reason before run

在 Trace A run 前標出 enter、hold、exit 與 service/J 方向。prediction 先於 baseline/candidate；policy 不能讀 future trace outcome。

B-01 : Trace A baseline

RUN THIS EXACTLY

POSIX / WSL ; A frozen predecessor active ;
package root

```
bash course.sh run --lab B --case trace-a-baseline  
# Windows: course.cmd run --lab B --case trace-a-baseline
```

Purpose: establish Trace A control.

Mechanism: A frozen predecessor → B baseline.

READ THE RECEIPT

fallback run ...*b2603b79* 、 energy *8.86 J* 、
delivered *4800 bit* ; fresh path 待重跑。

IF THE GATE IS RED

保留錯誤 ; restore A/B checkpoint 或
matching fallback ; 不可 retune 。

VERIFIED LOCAL FALLBACK | SIMULATED | fresh receipt/path pending

B-02 : 改 B marked block 並 freeze

RUN THIS EXACTLY

POSIX / WSL ; A frozen predecessor ; 只改
lab-b-enter-exit-hold
git diff -- student_policy.py
bash course.sh run --lab B --case trace-a-candidate
--freeze
Windows: course.cmd run --lab B --case trace-a-
Purpose: freeze B candidate.
Mechanism: B SHA + Trace A + predecessor.

READ THE RECEIPT

candidate policy c00ed808...78293718
freeze 2a4167e2...171349e0e
delivered 9600 bit ; fresh receipt pending

IF THE GATE IS RED

保留錯誤 ; restore A/B checkpoint 或
matching fallback ; 不可 retune 。

VERIFIED LOCAL FALLBACK | SIMULATED | fresh receipt/path pending

B-03 : 讀 transition 與 service ，不只讀品質

Context: trace-a-candidate-B result / endpoint replay

Run: rg -n 'quality_band|POLICY_DECISION|PACKET_ATTEMPT|delivered|expired|service_pass'
fallback_artifacts/trace-a-candidate-B/result.json

B CANDIDATE PACKET LEDGER

attempted 4
retransmissions 1
delivered 9600 bit
expired 1
energy 10.66 J

SERVICE VERDICT

service_pass = false
deadline_pass = false
先說 trade-off ，不說節能。

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

B-04 : 確認 freeze receipt 才能進 Trace B

Context: package root ; B frozen receipt / checkpoint
Run: python -m json.tool fallback_artifacts/receipts/b.json
find artifacts/checkpoints -maxdepth 1 -type f -print

baseline

candidate

lab-b-frozen

policy SHA · predecessor · active_block_id · scenario · anchor · seed · receipt SHA

FREEZE GATE | missing receipt / checkpoint = stop before withheld

B-05 : Trace B 不 retune

Context: POSIX / WSL ; B frozen policy active ; package root
Run: bash course.sh run --lab B --case trace-b
Windows: course.cmd run --lab B --case trace-b

FROZEN POLICY

policy SHA unchanged
no new freeze
read result + replay

TRACE B TRACE

new condition
queue / service / state / J
qualified verdict

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

B-06 : 辨識 too-slow 與 ping-pong

Context: trace-a-candidate-B vs trace-b-B ; same B policy SHA

Run: rg -n 'policy_sha256|service_pass|endpoint_energy_j|expired_packets'
fallback_artifacts/trace-b-B/result.json

~~python mason tool fallback_artifacts/trace-b-B/endpoint-replexicon~~

TOO-SLOW

hold 太久
錯過 service window
service loss

PING-PONG

enter ↔ exit 太近
多次 transition / retry
energy trade-off

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

Lab B debrief : hysteresis 是條件，不是保證

Context: Trace A / Trace B ; B receipt / policy / result pair

Run: `python -m json.tool fallback_artifacts/trace-a-candidate-B/result.json`

`python -m json.tool fallback_artifacts/trace-b-B/result.json`

condition

mechanism

evidence

verdict

在 Trace A / primary... ; 在 Trace B / hidden... ; 因此這個結論成立、被推翻，或待查。

FALLBACK_ARTIFACT | SIMULATED | NOT LIVE / NOT MEASURED

Recovery checkpoint : 保存 A/B lineage

Context: package root ; A/B receipt 、 checkpoint 、 result/replay pair
Run: find artifacts/checkpoints artifacts/receipts -maxdepth 1 -type f -print
find fallback_artifacts -maxdepth 2 -type f -name 'result.json' -print

A frozen

B frozen

result ↔ replay

prediction

status → exact role → restore ; 不覆寫既有 artifact , 不補假的 receipt 。